

THIRUVENGADAM S

+91 9790004113

thiruvengadamsamon@gmail.com

[LinkedIn](#)

[Google Scholar](#)

Chennai, India

EDUCATION

PGP Data Science & Engineering
Great lakes, Chennai

2018 – 2019

B.E Mechanical Engg.

Anna University, Chennai

2012 – 2016 | CGPA: 7.39

INTERESTS

Generative AI

NLP

Industrial AI

Mobility

SKILLS

PyTorch

Transformers

LangChain

Scikit-Learn

CV

Gradio

NumPy

Pandas

Langgraph

Python

AWS

Docker

Multi-Agent Systems

GIT

Linux

Cloud

Cuda

COURSEWORK

- Probability & Optimization
- Advanced Machine Learning & Deep Learning(Coursera)
- MLOps - Model Deployment & Production
- Cloud Computing & DevOps
- Six Sigma Green Belt (LIBA, Chennai)

AWARDS

- HIL Hackathon Winner 2020

WORK EXPERIENCE

R&D Center, Hitachi India Ltd. (HIL)

Bengaluru, India

Role: Build use cases | Problem hypothesis | Novel technology solutions | PoC

Staff AI Researcher

2025 – present

Sr. AI Research Engineer

2023 – 2025

AI Research Engineer

2020 – 2023

Industrial Generative AI

- Developed a novel image enhancement and chain of thoughts technique for **Vision LLMs** (Claude 3.5, GPT4o) to improve the accuracy of interpreting the **engineering diagrams** by 50%. [Patent filing][ICCV 2025 Paper]
- Developed multi-agent RAG systems utilizing vector databases, embedding models, retrieval techniques, and agent tools for rail maintenance, fault tree analysis - Manufacturing, ESG reporting, Smart Agri.

Mobility

- Developed Time-Aware SAITS imputation reducing sensor costs, achieving 9.5% lower Mean Relative Error. (cost saving) [Patent under filing]
- Built AEI-DRL adaptive ensemble imputation system reducing transportation occupancy MSE by 5.3% using reinforcement learning.

NLP (Text, Speech)

- Developed the **data engineering** pipeline to clean, pad, feature transform and structure Indian language raw speech data for speech model training.
- Trained **Automatic Speech Recognition (ASR)** models for a payment app in five Indian languages to promote inclusive banking. (WER < 5%)
- Finetuned **Speaker Verification models (ASV)** to improve accuracy on Indian languages, ensuring a 'secure user experience' for the payment app.
- Deployed speech AI models on **TorchServe** and on **Android OS** to assess latency, ensuring 'frictionless user experience' of the payment app.

Quberoot Analytics

Bengaluru, India

Data Scientist

2019 – 2020

- Developed real-time sentiment analysis using **BERT** for insurance sales, increasing conversion rates by 14% through dynamic optimization.
- Built semantic search web crawler using **Word2Vec** embeddings, reducing infrastructure RFP discovery time by 60%.

Lucas TVS Limited

Chennai, India

Graduate Trainee Engineer

2016 – 2018

- Implemented analytics dashboards across 7 plants, reducing production planning cycle time through value stream mapping optimization.

PUBLICATIONS

Journal & Conference: 4 | Patent (published): 2

- Adaptive Ensemble Imputation for Trip Data using Deep Reinforcement Learning (AEI-DRL), *2024 IEEE Big Data*.
- TA-SAITS: Time Aware-Self Attention based Imputation of Time Series algorithm for Partially Observable Multi-Variate Time Series, *ICMLA 2023*.
- Comparison of Different Speech Features for Connected Number Recognition of Indian Vernacular Languages, *SIGNAL 2023*.